Urban Building Energy Optimization

A Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY in
Computer Science and Engineering

Submitted by Anuj Kumar singh
(Roll No: 25scs1003005134) Under the supervision of Dr. Lalita Choudhary
Assistant Professor

IILM University, Greater Noida, Uttar Pradesh

(November2025)

Abstract

The goal of this project is to create a machine learning and Internet of Things-based system that optimizes household energy use in urban settings. To forecast short-term energy demand, the system combines sophisticated machine learning models with simulated IoT sensor data, including temperature, humidity, occupancy, and appliance usage. The project analyzes daily and hourly consumption patterns, identifies key factors influencing energy usage, and produces visual insights through trend analysis and graphs using algorithms like Random Forest and Linear Regression. The forecasting results show that prediction powered by machine learning greatly improves energy management efficiency. The suggested framework can be expanded to create an automated, real-time smart home system that would save money, cut down on waste, and increase sustainability

Keywords: -

IoT, Machine Learning, Energy Optimization, Forecasting, Smart Home, Urban Energy Consumption, Random Forest, Energy Prediction, Data Analytics

Contents

1. Introduction. 5

1.1 Overview 5

1.2 Objectives 5

1.3 Problem Statement 6

2. Literature Review. 6

3.Methodology. 7

3.1 System Architecture. 7

3.2 Data Collection. 8

3.3 Preprocessing. 8

3.4 Model Training & Evaluation. 8

4. Software Tools 8

4.1 Python Programming Language 8

4.2 Python Libraries 9

5. Implementation & Result 9-16

5.1 Dataset Creation 16

5.2 Data Preprocessing . . . . . … 16

5.3 Model Training 17

5.4 Model Testing & Evaluation 17

5.5 Visualizations 17

5.6 Final Output 17

6. Results and Discussion 18

6.1 Model Performance 18

6.2 Visual Analysis 18

6.3 Discussion 18

7. Conclusion 19

8. Future Scope 19

9. References 19

1. Introduction

The demand for energy resources has increased dramatically due to the exponential rise of urban regions. Urban Over two-thirds of the world's energy consumption occurs in these regions, and a significant portion of it comes from the energy of homes and business buildings. Building energy management systems are typically made to operate on fixed schedules and lack flexibility, which leads to ineffectiveness. Using advancements in AI/ML, smart sensors, and the integration of renewable energy, the potential for creating intelligent, self-optimizing structures that can alter function is fulfilled in accordance with occupancy and environmental conditions. The goal of the research is to provide an inclusive energy consumption in urban structures comprising management, optimization, and forecasting.

1.1 Overview

This project uses machine learning and the Internet of Things to create a smart home energy management system. To identify trends in energy consumption, IoT-based sensor data—such as temperature, humidity, occupancy, and appliance usage—is examined. Short-term energy consumption is predicted using machine-learning models such as XGBoost, Random Forest, and Linear Regression. The information aids in identifying important variables influencing consumption and suggests methods to cut waste. Overall, the experiment demonstrates how integrating ML forecasting with IoT monitoring can improve the sustainability and efficiency of household energy use.

1.2 Objectives

● To create an IoT-based system for collecting household data like temperature, humidity, occupancy, and appliance usage.
● To clean and analyze the data to understand energy usage patterns.
● To build and compare ML models (Linear Regression, Random Forest, XGBoost) for predicting short-term energy consumption.
● To evaluate model accuracy using MSE, R^2 Score, and visual graphs.
● To suggest energy-saving strategies based on prediction results.
● To develop a system that can be expanded into a real-time smart home energy management framework

1.2 Problem Statement

Due to an increase in equipment, erratic consumption patterns, and a lack of real-time monitoring, urban families frequently experience rising energy usage. Conventional energy management techniques don't adjust to shifting occupancy and environmental circumstances since they rely on human operation and set timetables. Significant energy is squandered as a result, which raises expenses and has an unnecessary negative impact on the environment. The majority of homes do not make good use of the detailed household data that IoT devices are able to gather. An intelligent system that can evaluate sensor data, forecast future energy usage, and offer insightful information for consumption optimization is required. Creating a dependable, data-driven system that combines machine learning predictions with Internet of Things monitoring to enable effective and sustainable energy management is the difficult part.

2. Literature Review

Since electricity consumption has been rising over time, a lot of research has concentrated on enhancing household energy use. Researchers have discovered that irregular appliance usage and inadequate monitoring result in significant energy waste. Because IoT devices can gather real-time data like temperature, humidity, and appliance activity, they are frequently recommended as a solution to this problem. Another topic that has received a lot of attention in the literature is machine learning. Previous research has employed various models, including XGBoost, Random Forest, and Linear Regression, to forecast energy usage. The majority of studies concur that machine learning models are capable of identifying trends in electricity consumption and producing reasonably accurate forecasts. Additionally, some studies demonstrate how integrating IoT data with machine learning techniques can assist in determining energy waste and ways to minimize it. This method can help with better home appliance control and more intelligent planning. Overall, prior research backs up the notion that machine learning forecasting and IoT-based data collection can complement one another to improve household energy efficiency. The same idea is used in this project, which predicts energy consumption using ML models and simulated IoT-like data.

3. Methodology

For this project, I mainly followed a simple step-by-step approach. First, I created a dataset that behaves kinda like it came from IoT sensors. Then I cleaned the data, tried out a few ML models, and checked which one gave the best predictions. The whole idea was basically to see if ML can actually predict household electricity use in a real-life kind of situation.

3.1 System Architecture

The system can be broken down into four small parts:

1. IoT Sensing Layer

   Since I didn't have actual sensors, I made some sample data that looks like what sensors would give.

8 It includes things like temperature, humidity, whether someone is home or not, and what appliances might be used, etc.

2. Data Processing Layer

Here I cleaned up the data. Basically, things like removing/filling missing values, creating new columns like hour or day to make the data more meaningful, and converting everything into a format the ML models can actually work with.

3. Machine Learning Layer

I used three models here:

- Linear Regression
- Random Forest
- XGBoost All three were trained to guess the energy usage for a given hour.

4. Visualization & Insights Layer

I plotted graphs to see how the energy usage changes and how accurate the predictions were. These graphs helped a lot in understanding what's going on instead of just looking at numbers.

3.2 Data Collection

I didn't have real IoT data, so I generated a fake/synthetic dataset. It tries to follow normal household behaviour and includes hourly usage, some weather-type values, and random variations that normally happen in real life.

3.3 Preprocessing

Before training the models, I did some basic prep work:

- fixed missing values
- created features like hour, day, month
- scaled some values so the models don't get confused

3.4 Model Training & Evaluation

I used 80% of the dataset to train the models and kept 20% for testing. To check how well the models performed

4. Software Tools

The following software tools and libraries were used to develop, analyze, and evaluate the proposed energy optimization system

4.1 Python Programming Language

Used as the primary development environment for data processing, visualization, and machine-learning model implementation

4.2 Python Libraries

- Pandas: For data handling, cleaning, and analysis.
- NumPy: For numerical computations and array operations.
- Matplotlib: Used to generate graphs and visualize trends.
- Seaborn: For advanced statistical visualizations.
- Scikit-Learn: Provides ML algorithms such as Linear Regression and Random Forest.
- XGBoost: Used for gradient boosting–based prediction with high accuracy.

● Warnings & Sys Modules: To manage system-based settings and suppress unnecessary warnings

5. Implementation

```
# ------------------------------------------------------------
# 🏠 Household Energy Analysis and Prediction (All Graphs Inline)
# ------------------------------------------------------------

import sys
sys.stdout.reconfigure(encoding='utf-8')  # ✅ Fix for Windows encoding issue

# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings('ignore')

# Set random seed for reproducibility
np.random.seed(42)

# ------------------------------
# 1. Create Simulated Household Energy Dataset
# ------------------------------
print("Creating simulated household energy dataset...")

num_households = 100
```

```
dates = pd.date_range(start='2025-09-01', end='2025-09-30 23:00:00', freq='H')
num_records = len(dates) * num_households // 10  # Subsample for manageability

household_ids = [f'House_{i+1}' for i in range(num_households)]
data = {
    'Timestamp': np.random.choice(dates, size=num_records),
    'Household_ID': np.random.choice(household_ids, size=num_records),
    'Temperature_C': np.random.normal(20, 5, size=num_records),
    'Humidity_%': np.random.uniform(30, 90, size=num_records),
    'House_Size_sqm': np.random.uniform(50, 200, size=num_records),
    'Num_Occupants': np.random.randint(1, 6, size=num_records),
    'Appliance_Usage': np.random.uniform(0.5, 5.0, size=num_records),
    'Energy_Consumption_kWh': np.zeros(num_records)
}

# Simulate energy consumption (dependent on features)
for i in range(num_records):
    base_energy = 0.1 * data['House_Size_sqm'][i] + 0.5 *
data['Num_Occupants'][i] + 0.8 * data['Appliance_Usage'][i]
    weather_effect = 0.05 * data['Temperature_C'][i] + 0.02 *
data['Humidity_%'][i]
    noise = np.random.normal(0, 0.5)
    data['Energy_Consumption_kWh'][i] = max(0, base_energy + weather_effect +
noise)

df = pd.DataFrame(data)
df['Timestamp'] = pd.to_datetime(df['Timestamp'])

print("✅ Dataset created successfully!\n")


#-----------------------------
# 2. Data Preprocessing
# -----------------------------
print("Preprocessing data...")

df.fillna(df.mean(numeric_only=True), inplace=True)
df['Hour'] = df['Timestamp'].dt.hour
df['Day_of_Week'] = df['Timestamp'].dt.dayofweek
df['Is_Weekend'] = df['Day_of_Week'].isin([5, 6]).astype(int)
df['Month'] = df['Timestamp'].dt.month

daily_energy = df.groupby([df['Timestamp'].dt.date,
'Household_ID'])['Energy_Consumption_kWh'].sum().reset_index()
daily_energy.rename(columns={'Timestamp': 'Date'}, inplace=True)
```

```python
print("\nSample of daily aggregated data:")
print(daily_energy.head())

# ------------------------------
# 3. Exploratory Data Analysis (EDA)
# ------------------------------
print("\nPerforming EDA...")

print("\nDataset Summary:")
print(df.describe())

# Store all figures to show together later
figures = []

# 🔹 Correlation Matrix
fig1 = plt.figure(figsize=(10, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm',
fmt='.2f')
plt.title('Correlation Matrix of Features')
figures.append(fig1)

# 🔹 Daily Energy Trend
daily_trend =
df.groupby(df['Timestamp'].dt.date)['Energy_Consumption_kWh'].mean()
fig2 = plt.figure(figsize=(12, 6))
daily_trend.plot(kind='line', marker='o', color='blue')
plt.title('Average Daily Energy Consumption (kWh)')
plt.xlabel('Date')
plt.ylabel('Energy Consumption (kWh)')
plt.xticks(rotation=45)
figures.append(fig2)


# 🔹 Energy by Hour
hourly_energy = df.groupby('Hour')['Energy_Consumption_kWh'].mean()
fig3 = plt.figure(figsize=(10, 6))
hourly_energy.plot(kind='bar', color='green')
plt.title('Average Energy Consumption by Hour')
plt.xlabel('Hour of Day')
plt.ylabel('Energy Consumption (kWh)')
figures.append(fig3)

# ------------------------------
# 4. Prepare Data for ML
# ------------------------------
print("\nPreparing data for ML...")
```

```
features = ['Temperature_C', 'Humidity_%', 'House_Size_sqm', 'Num_Occupants',
'Appliance_Usage', 'Hour', 'Is_Weekend']
X = df[features]
y = df['Energy_Consumption_kWh']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# ------------------------------
# 5. Train and Evaluate ML Models
# ------------------------------
print("\nTraining and evaluating ML models...")

models = {
    'Linear Regression': LinearRegression(),
    'Random Forest': RandomForestRegressor(n_estimators=100, random_state=42),
    'XGBoost': XGBRegressor(n_estimators=100, random_state=42)
}

results = []

for name, model in models.items():
    model.fit(X_train_scaled, y_train)
    y_pred = model.predict(X_test_scaled)

    mse = mean_squared_error(y_test, y_pred)
    r2 = r2_score(y_test, y_pred)
    cv_scores = cross_val_score(model, X_train_scaled, y_train, cv=5,
scoring='r2')

    results.append({
        'Model': name,
        'MSE': mse,
        'R2': r2,
        'CV_R2_Mean': cv_scores.mean(),
        'CV_R2_Std': cv_scores.std()
    })

    # Plot Actual vs Predicted
    fig_model = plt.figure(figsize=(8, 6))
    plt.scatter(y_test, y_pred, alpha=0.5)
```

```
    plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
'r--')
    plt.title(f'{name}: Actual vs Predicted Energy Consumption')
    plt.xlabel('Actual (kWh)')
    plt.ylabel('Predicted (kWh)')
    figures.append(fig_model)

results_df = pd.DataFrame(results)
print("\nModel Performance:")
print(results_df)

# ------------------------------
# 6. Feature Importance (Random Forest)
# ------------------------------
print("\nAnalyzing feature importance...")

rf_model = models['Random Forest']
feature_importance = pd.DataFrame({
    'Feature': features,
    'Importance': rf_model.feature_importances_
}).sort_values(by='Importance', ascending=False)

fig_imp = plt.figure(figsize=(10, 6))
sns.barplot(x='Importance', y='Feature', data=feature_importance)
plt.title('Feature Importance for Random Forest')
figures.append(fig_imp)

print("\nFeature Importance:")
print(feature_importance)









# ------------------------------
# 7. Predict Future Energy Consumption
# ------------------------------
print("\nPredicting future energy consumption...")

future_dates = pd.date_range(start='2025-10-01', end='2025-10-07 23:00:00',
freq='H')
future_data = {
    'Timestamp': future_dates,
    'Temperature_C': np.random.normal(20, 5, len(future_dates)),
```

```
    'Humidity_%': np.random.uniform(30, 90, len(future_dates)),
    'House_Size_sqm': np.random.choice(df['House_Size_sqm'],
len(future_dates)),
    'Num_Occupants': np.random.choice(df['Num_Occupants'], len(future_dates)),
    'Appliance_Usage': np.random.uniform(0.5, 5.0, len(future_dates)),
    'Hour': pd.to_datetime(future_dates).hour,
    'Is_Weekend': pd.to_datetime(future_dates).dayofweek.isin([5,
6]).astype(int)
}
df_future = pd.DataFrame(future_data)

X_future = df_future[features]
X_future_scaled = scaler.transform(X_future)
future_predictions = rf_model.predict(X_future_scaled)
df_future['Predicted_Energy_kWh'] = future_predictions

fig_future = plt.figure(figsize=(12, 6))
plt.plot(df_future['Timestamp'], df_future['Predicted_Energy_kWh'],
label='Predicted Energy (kWh)', color='purple')
plt.title('Predicted Household Energy Consumption (Next 7 Days)')
plt.xlabel('Timestamp')
plt.ylabel('Energy Consumption (kWh)')
plt.xticks(rotation=45)
plt.legend()
figures.append(fig_future)

# ----------------------------
# 8. Show all graphs at once
# ----------------------------
print("\n📊 Displaying all plots together...")
for fig in figures:
    plt.figure(fig.number)
plt.show()

print("\n✅ Project completed successfully! All graphs are displayed
together.")
```

Result

```
Creating simulated household energy dataset...
✅ Dataset created successfully!

Preprocessing data...

Sample of daily aggregated data:
         Date Household_ID  Energy_Consumption_kWh
0  2025-09-01      House_1               24.759839
1  2025-09-01     House_10               17.341545
2  2025-09-01    House_100               16.721807
3  2025-09-01     House_11               24.363829
4  2025-09-01     House_12               87.324109

Performing EDA...

Dataset Summary:
                      Timestamp  Temperature_C  ...   Is_Weekend   Month
count                      7200    7200.000000  ...  7200.000000  7200.0
mean   2025-09-15 23:58:58.500000      19.980789  ...     0.266250     9.0
min           2025-09-01 00:00:00       2.871023  ...     0.000000     9.0
25%           2025-09-08 11:00:00      16.587557  ...     0.000000     9.0
50%           2025-09-16 02:00:00      19.979224  ...     0.000000     9.0
75%           2025-09-23 13:00:00      23.359073  ...     1.000000     9.0
max           2025-09-30 23:00:00      37.093061  ...     1.000000     9.0
std                           NaN       4.988480  ...     0.442027     0.0

[8 rows x 11 columns]

Preparing data for ML...

Training and evaluating ML models...

Model Performance:
               Model       MSE        R2  CV_R2_Mean  CV_R2_Std
0  Linear Regression  0.251424  0.987600    0.987457   0.000255
1      Random Forest  0.354603  0.982511    0.980984   0.000941
2            XGBoost  0.368038  0.981848    0.982011   0.000817

Analyzing feature importance...
```

```
Feature Importance:
           Feature  Importance
2   House_Size_sqm    0.909249
4  Appliance_Usage    0.056373
3    Num_Occupants    0.018755
1       Humidity_%    0.007721
0    Temperature_C    0.005319
5             Hour    0.002198
6       Is_Weekend    0.000385

Predicting future energy consumption...

📊 Displaying all plots together...

✅ Project completed successfully! All graphs are displayed together.
```

5.1 Dataset Creation

This dataset includes hourly values for:

- Temperature
- Humidity
- Occupancy
- Appliance usage
- Total energy consumption

The data was designed to behave like real household readings so that the models could learn actual usage patterns.

5.2 Data Preprocessing

Before training the models, the dataset was cleaned and prepared:

- Missing values were filled
- New time-based features (Hour, Day, Month) were created
- Numerical values were scaled
- The dataset was split into training (80%) and testing (20%)

5.3 Model Training

Three machine-learning models were implemented:

1. Linear Regression
2. Random Forest Regressor

5.3 Model Testing & Evaluation

The trained models were tested using the 20% testing data. Their performance was evaluated using:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score
- Actual vs Predicted graphs

This helped compare how accurate each model was in predicting energy consumption.

5.4 Visualizations

Graphical outputs were created using Matplotlib to analyze:

- Energy usage trends
- Temperature vs Energy relationship
- Feature importance
- Model predictions vs actual values

5.5 Final Output

Based on testing, XGBoost gave the best accuracy among all models. The system successfully showed how IoT-like data and ML can be combined to forecast household energy consumption.

5.4 Model Testing & Evaluation

The trained models were tested using the 20% testing data. Their performance was evaluated using:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score
- Actual vs Predicted graphs

This helped compare how accurate each model was in predicting energy consumption.

5.6 Visualizations

Graphical outputs were created using Matplotlib to analyze:

- Energy usage trends
- Temperature vs Energy relationship
- Feature importance
- Model predictions vs actual values

5.7 Final Output

Based on testing, XGBoost gave the best accuracy among all models. The system successfully showed how IoT-like data and ML can be combined to forecast household energy consumption.

6. Results and Discussion

6.1 Model Performance

Three machine-learning models were trained to predict household energy consumption:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor

6.2 Visual Analysis

1. Energy Trend Graph
2. Temperature vs Energy Scatter Plot
3. Actual vs Predicted Graphs
4. Feature Importance Plot Important factors identified:

- Temperature
- Appliance usage
- Occupancy
- Hour of the day

6.3 Discussion

The analysis shows that machine learning is actually quite useful for predicting how much electricity a household will use. The models were able to pick up different usage patterns and understand how factors like temperature and occupancy affect energy consumption. Because of this, the predictions turned out to be quite accurate.

These kinds of forecasts can help families avoid unnecessary electricity wastage, plan when to use appliances, and even lower their monthly bills. Overall, the system clearly shows how IoT devices and machine learning can work together to support smarter and more automated home energy management in the future.

6. Conclusion

In this project, IoT-like data was used to predict household energy consumption using machine learning. The models demonstrated that energy demand could be reasonably predicted and were able to learn various usage patterns. This

aids in determining where electricity may be wasted and when it is being used more. All things considered, the project demonstrates how ML combined with IoT-style data can facilitate more intelligent and effective home energy management.

7. Future Scope

There are several ways this work can be improved in the future:

- Real IoT sensors can be used instead of simulated data to make the system more accurate.
- A real-time energy monitoring and control system can be developed to automatically adjust appliances.
- More advanced models, like LSTM or deep learning networks, can be added for better predictions.
- A mobile or web dashboard can be created so users can easily see their energy usage.
- The system can be expanded to support multiple households or even small communities.